Physical Chemistry cume

April 29th 2016

1) What are all the possible quantum numbers for an electron in (a) the 3d orbital, (b) the 2p, (c) the 2s, and (d) the 4s orbital?
(30 points)

2) (a) The normalized radial wavefunction for the 2s orbital is $R_{2,0} = \frac{1}{\sqrt{8}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2 - \frac{Z \cdot r}{a_0}\right)e^{-\frac{zr}{2a_0}}$

Where:
Z is the nuclear charge (i.e. number of protons in the nucleus)
a_0 is the Bohr radius = 52.9 pm

Find the average (i.e. expectation value of) distance of the electron from the nucleus for Li^{2+} using the postulates of quantum mechanics.

(40 points)

(b) Show that the same result is obtained using the following equation:

$$\langle r \rangle_{n,l} = \frac{n^2 a_0}{Z}\left[1 + \frac{1}{2}\left(1 - \frac{l(l+1)}{n^2}\right)\right]$$

(15 points)

(c) Using the equation from question 2(b), determine the average distance of an electron in the $2p_z$ orbital for Li^{2+}.
(15 points)

You may find the following information useful:

$$\langle a \rangle = \frac{\int \Psi^* \hat{A} \Psi d\tau}{\int \Psi^* \Psi d\tau}$$

HINT: If normalized: $\int \Psi^* \Psi d\tau = 1$

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{(n+1)}}$$